Module - 02

* Block Coding :-

→ In block coding, message is divided into blocks, each of k bits, called datawords.

→ Add r redundant bits to each block to make length $n = k + r$. The resulting n-bit blocks are called codewords

→ With k bits, 2^k datawords can be created.

→ With n bits, 2^n codewords can be created.

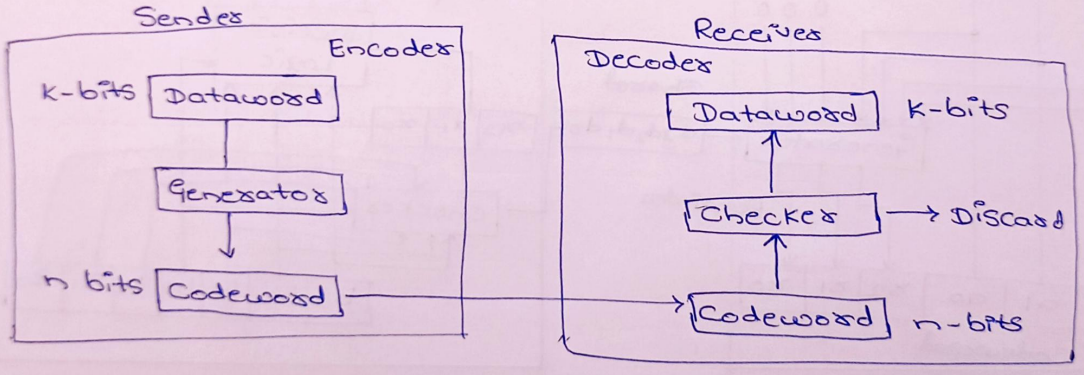
→ Since $n > k$, the possible codewords is larger than number of possible datawords

* Error Detection :-

→ If the following two conditions are met, the receiver can detect a change in the original codeword.

① The receiver has a list of valid codewords

② The original codeword has changed to an invalid one



* Hamming Distance :-

→ The Hamming distance between two words (of same size) is the number of differences between corresponding bits.

→ The Hamming distance between two words x & y is shown as d(x,y)

→ The Hamming distance can be easily be found by applying the XOR operation on the two words & count number of 1s in the result.

→ Ex :- d(10101, 11110) is 3

* Cyclic Codes :-

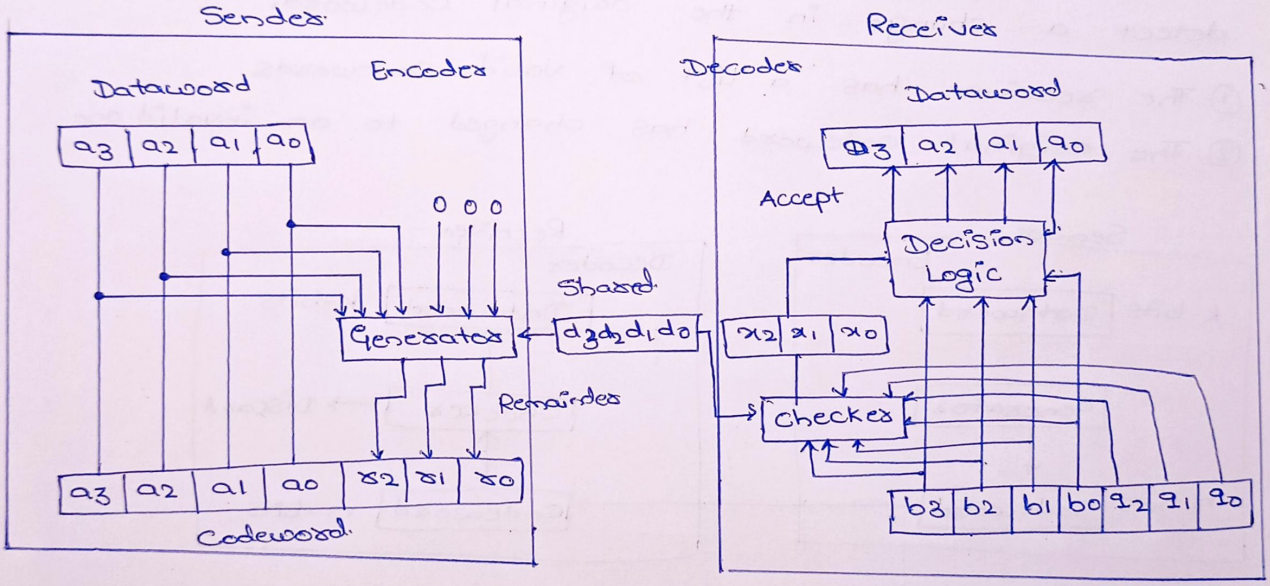
→ Cyclic codes are special linear block codes with one extra property.

→ In a cyclic code, if a codeword is cyclically shifted (rotated), the result is another codeword.

→ Ex: If 1011000 is a codeword & we cyclically shift then 0110001 is also codeword.

* Cyclic Redundancy check :-

→ It is a subset of cyclic codes called Cyclic redundancy (CRC), which is used in networks such as LANs & WANs



* Encoder :-

→ In the encoder, the dataword has k bits (4 bits), code word has n bits (7 here).

→ The data word is augmented by adding n-k 0s to the right hand side of the word.

→ The n-bit result is fed to the generator.

→ The generator uses a divisor of size n-k+1 which is predefined and agreed upon.

→ The generator divides the augmented dataword by divisor

→ The quotient of the division is discarded.

* Decoder:-

→ The decoder receives the corrupted codeword.

→ A copy of all n bits are fed to the checker.

→ A remainder produced by checker is a Syndrome of n-k bits, which is fed to the decision logic analyzer.

→ If Syndrome bits are 0, the 4 left most bits of the code word are accepted as a data word.

* Problems:- Encoder:-

→ Dataword : 1001

Divisor : 1011

```
      1010
  ------------
1011 | 1001000
     | 1011
     -------
      0100
      0000
      ----
       1000
       1011
       ----
        0110
        0000
        ----
         110
```

Codeword = Dataword + remainder

= 1001 + 110

= 1001110

Decoder

```
      1010
  ------------
1011 | 1001110
     | 1011
     -------
      0101
      0000
      ----
       1011
       1011
       ----
        0000
        0000
        ----
         000
```

Dataword accepted.

* Media Access Control :-

→ When stations are connected to a common link (multipoint), multiple access protocol is required to coordinate access to the link.

→ In random access no station is superior to another station & non is assigned control over another.

* Two features :-

→ There is no scheduled time for the station to transmit.

→ Transmission is random among the station.

→ No rules specify which station should send next. Stations compete with one another to access the medium.

1) Pure ALOHA :-

→ All stations can transmit data whenever they have data ready to be sent.

→ Every station transmit data without checking whether the channel is free or not.

→ After sending a frame, the sender expects acknowledgement from the receiver.

→ If acknowledgement doesn't arrive in time, the sender waits for random time and resend the frame again.

→ waiting time for each station is different.

→ Vulnerable time for pure Aloha = $2 \times Tt$

→ In Pure Aloha, the probability of successful transmission of the data packet. = $e_1 \times e^{-2g}$

→ In Pure Aloha max efficiency = 18.4%.

2) Slotted ALOHA :-

→ Slotted ALOHA improves efficiency of pure ALOHA.

→ In this, any station can transmit the data at the beginning of any time slot.

→ Divides time into discrete intervals equivalent to length of the frame.

→ Vulnerable time for slotted Aloha = Tt

→ In Slotted Aloha, the probability of successful transmission of the data packet. $= q * e^{-q}$

→ In Slotted Aloha, Max efficiency = 36.8 %.

3) CSMA :-

→ Carrier Sense Multiple Access.

→ To minimize the chance of collision and, therefore, increase the performance, the CSMA method was developed.

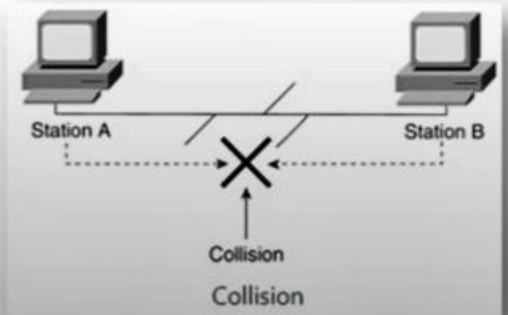
→ The chance of collision can be reduced if a station senses the medium before trying to use it.

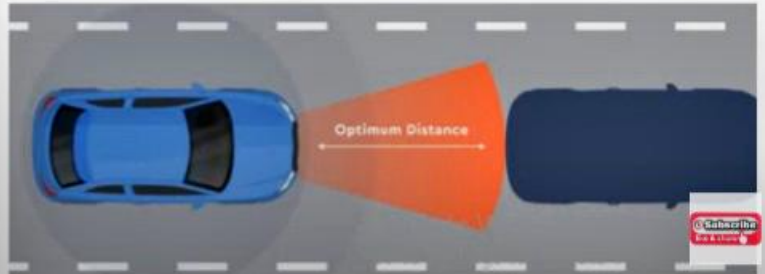
→ CSMA requires that each station first listen to the medium before sending

→ CSMA is based on principle "Sense before transmit" or "listen before talk".

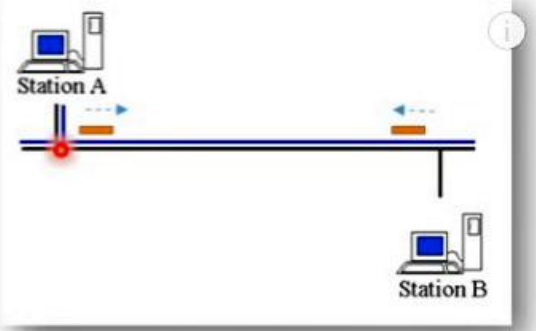
CSMA / CA

- CSMA stands for Carrier Sense Multiple Access with Collision Avoidance.
- It is similar to the CSMA CD protocol that operates in the media access control layer.
- It avoid a collision in network rather than allowing it to occur.
- When station sends a data frame to a channel, it checks whether channel is busy or idle.
- If the shared channel is busy, the station waits until the channel enters idle mode.
- It reduces the chances of collisions.
- Due to this medium is send data packets more efficiently.





CSMA / CD



- It stands for Carrier Sense Multiple Access / Collision Detection.

➢Working:

1. **Carrier Sensing:** Station first check that network is idle or busy. If it is busy it wait until it become idle.
2. **Data Transmission:** When the network is idle, the device begins transmitting its data packet.
3. **Collision Detection:** Device continues to monitor network. If it detects that another device has transmitted data at the same time it stops transmitting immediately.
4. **Collision Handling:** Device sends special signal known as "jam signal" to inform all other devices that collision has occurred so other devices waits for a random amount of time.
5. **Retransmission:** Device will try to transmit its data again, starting with carrier sensing to ensure the network is idle.



Activate Windows

* Flow - Control :- STOP & WAIT ARQ

→ ARQ (Automatic Repeat Request)

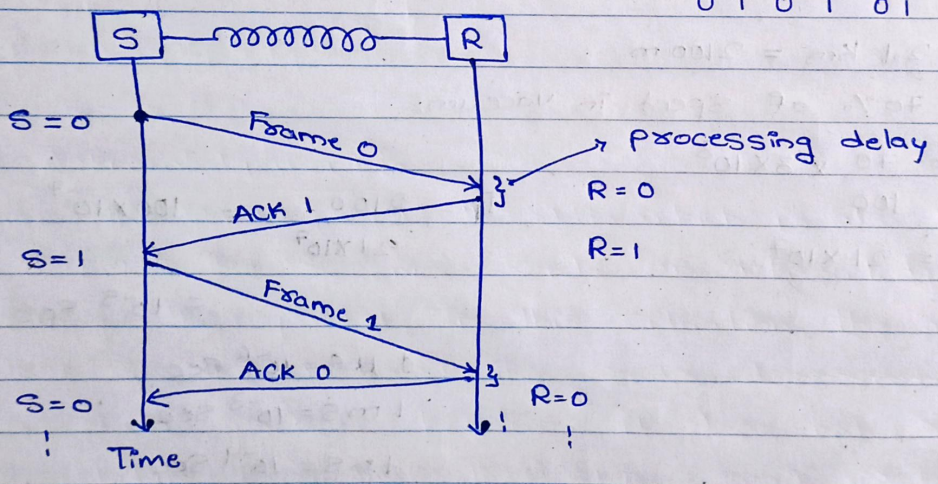
→ The stop and wait ARQ protocol sends a data frame and then waits for an acknowledgement (ACK) from the receiver.

→ The ACK indicates that the receiver successfully received the data frame.

→ After receiving the ACK from the receiver, the Sender delivers the next data frame.

→ So there is a stop before the next data frame is transferred, hence it is known as the stop and wait ARQ protocol

→ It uses link b/w Sender & receiver as a half-duplex link.



for L-bits

$L \longrightarrow T_t + T_p + T_{pr} + T_t^a + T_p$

$L \longrightarrow T_t + 2T_p$ Sec

Total Cycle :- $T_t + T_p + T_p$ (ACK ↑, Sender to Receiver ↑)

η = efficiency = $\frac{\text{useful Time}}{\text{Total time}} = \frac{T_t}{(T_t + 2T_p)}$

$a = \frac{T_p}{T_t}$

$$= \frac{T_t}{1 + 2\frac{T_p}{T_t}}$$

$$\eta = \frac{T_t}{1 + 2a}$$

* At Sender :-

1) Send one data Packet at a time.
2) Send the next Packet only after receiving acknowledgment for the previous.

* At Receiver :-

1) Send the acknowledgment after receiving and consuming data Packet.
2) After consuming packet acknowledgment need to be Sent.

* Throughput / Bandwidth - utilization / Effective Bandwidth :-

↓ bps

$$\Rightarrow \frac{L}{T_t + 2T_p} = \eta : B$$

↳ Bandwidth of the medium

$$= \frac{\frac{L}{B} \times B}{T_t + 2T_p} = \frac{T_t \times B}{T_t + 2T_p}$$

$$= \eta \times B$$

1) Go-back-N (Sliding Window):

* Sender Side:

→ The Sender maintains a window of frames that can be sent without waiting for an ACK

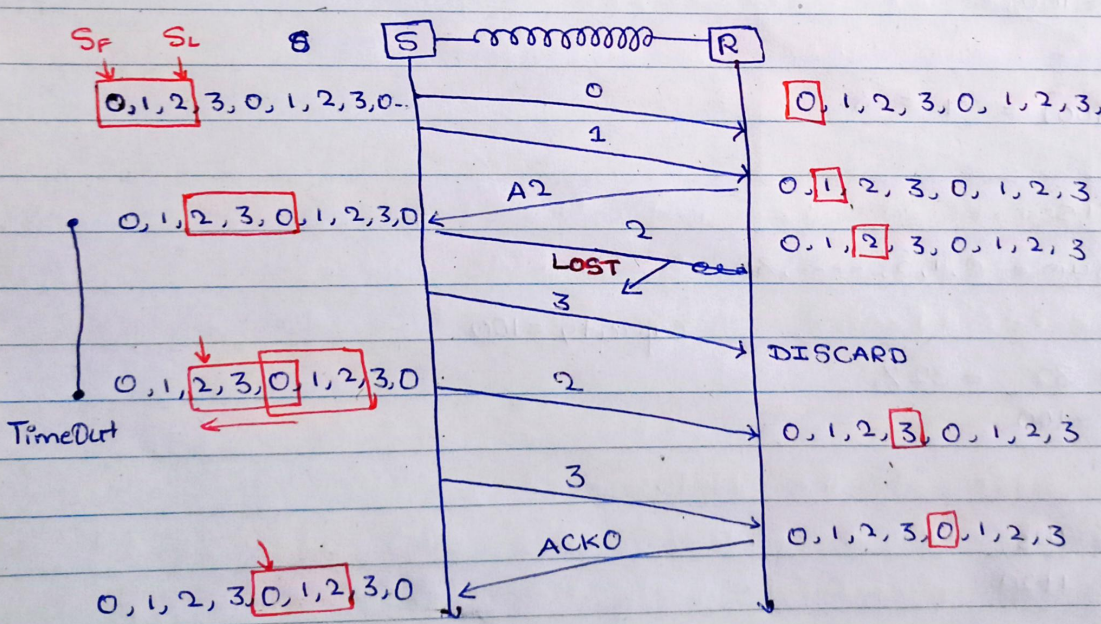
→ Frames are sent sequentially, and the Sender waits for ACKs

→ If an ACK is not received within the timeout period, the Sender retransmits all frames starting from lost frame.

* Receiver Side:

→ The receiver accepts frames in sequence & sends an ACK for the last correctly received frame.

→ If a frame is lost or corrupted, the receiver discards all subsequent frames & does not send an ACK for them.



→ min no. of Seq num needed in Stop & wait = 2 = $W_S + W_R$

→ min no. of Seq num needed in GBN = $W_S + 1$

$W_S \leq 2^m - 1$

2^m = # Seq num

* Problems :- Pure ALOHA

Q) A pure ALOHA network transmits 200-bit frames on a shared channel of 200 Kbps. What is the throughput if the system (all stations) produces 1000 frames per second?

→ * Frame Transmission rate :-

$$FTR = \frac{\text{Frame Size}}{\text{Channel Bandwidth}} = \frac{200}{200 \times 1000} = 0.001 \text{ Sec} = 1 \text{ ms}$$

* If the system creates 1000 frames per second, then 1 frame per millisecond.

Therefore $G = 1$. (G = Given frames per second x frame transmission rate)

Throughput $S = G \times e^{-2G}$

$= 1 \times e^{-2(1)}$

$= 1 \times e^{-2}$

$= 0.1353$

$= 13.53\ \%$.

→ This means that the throughput is $1000 \times 0.135 = 135$ frames

→ only 135 frames out of 1000 will probably survive.

(2) For 500 frames per second.

→

$$FTR = \frac{200}{200 \times 1000} = 1 \text{ ms}$$

$$G = 500 \times 0.001 = 0.5 = \frac{1}{2}$$

Throughput $S = G \times e^{-2G}$

$= 0.5 \times e^{-2(1/2)}$

$= e^{-1} \times 0.5$

$= 0.1839$

$= 18.39\%$

→ This means throughput is $500 \times 0.184 = 92$

* Problems on Slotted ALOHA :-

Q) A Slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth Find the throughput if the System produces 1000 frames per second?

→ * $FTR = \frac{200}{200 \times 1000} = 10^{-3} S = 1ms$

$G = 1000 \times 0.001 = 1$

Throughput $= G \times e^{-G}$

$= 1 \times e^{-1}$

$= 0.368.$

This means that throughput is $1000 \times 0.368 = 368$ frames

→ only 368 out of 1000 frames will probably Survive.

(2) For 250 frames :-

* $FTR = \frac{200}{200 \times 1000} = 10^{-3} S = 1ms$

$G = 250 \times 0.001 = 0.25$

Throughput $= G \times e^{-G}$

$= 0.25 \times e^{-0.25}$

$= 0.25 \times 0.779$

$= 0.195$

This Means throughput is $250 \times 0.195 = 49$ frames

Only 49 frames out of 250 will probably Survive.